

$$\mu' = \mu + \phi'^2 \sum_{j=1}^m g(\phi_j) (s_j - E(\mu, \mu_j, \phi_j)) \quad (7)$$

Overall, the Glicko-2 rating update is applied after each game, while our work tackles the challenge of rating updates after each move within a game.

3 Related Works

One of the first works on modeling chess player skill used Rybka 3 [31] evaluations to model player moves based off rating [22]. Later, 30 hand-crafted features were fed into random forests [1] and support vector machines [4] to predict ratings on players binned by rating into 10 different splits [28]. Rather than focusing the problem as a classification problem with limited bins, we choose to frame it as a regression problem, as this gives us more flexibility and precision in rating players. Also, the data from the previous rating estimation paper is not publicly available, so we compiled our own open source dataset from Lichess for this task.

The recent popularity of word prediction has permeated its way into the chess field as move prediction. Maia [17] adapted AlphaZero’s residual Convolutional Neural Network (CNN) [26], removed the search algorithm to predict moves based on rating bins, and improved the accuracy over a depth-limited Stockfish. Stockfish is currently the strongest chess engine [29] and the original Leela Zero was an open source clone of it, although it has since migrated to a transformer architecture [19]. The Maia model was then fine-tuned to predict specific player moves [18]. GPT-2 was fine-tuned on 2.8 million chess games, demonstrating strategic understanding without directly obtaining a rating. Another transformer model achieved a 2895 Lichess Blitz rating through supervised learning on Stockfish evaluations from 10 million games [24].

Further classification of chess games has been explored through a binary classifier for detecting cheaters, using the Euclidean distance to compare player features such as move accuracy and response time [12]. One paper showed that the winning likelihood strongly correlates with the remaining time in rapid chess games, essentially less time lowers chances of winning [25]. Multi-class classification of chess games, predicting outcomes such as win/draw/loss based on moves and ratings with an LSTM, has also been attempted [21]. Another study predicted the brilliance of moves (a binary classification) using features from chess engine analysis [33]. While these approaches are similar to ours, they focus on classification tasks, whereas our model predicts a continuous value.

Rating prediction has been conducted in other gaming domains, such as Massive Online Battle Arena (MOBA) games. In one case, a transformer was trained to predict player ratings for matchmaking using descriptive features like kills and deaths recorded in three-minute intervals [34]. The predicted rating showed a higher correlation with the win rate than the existing rating system, particularly for players with fewer than 15 games.

4 Dataset

1.2 million Lichess games from April 2021 to July 2024 are used in the dataset, with 30,000 games from each month.¹ Lichess is a chess website chosen because it is open source and contains the largest publicly available database of games for free use. The data is split randomly with 80% used for the training set and 20% used for the test set. Other available input features include the amount of clock time remaining after each move. The outputs to be estimated are the ratings of both players for a single game.

Moreover, we applied the same model architecture to the 2024 IEEE Big Data 2024 Cup Chess Puzzle Difficulty Competition [11]. This data contains a Forsyth-Edwards Notation (FEN) [5] string representation of the starting position, along with the moves in Universal Chess Interface (UCI) [10] notation for the solution.

4.1 Time Controls

Lichess time controls are less than 29 seconds for UltraBullet, 179 for Bullet, 479 for Blitz, 1499 for Rapid, else Classical. Games with increment add 40 times the increment to the total time [15]. A time control of "5+3", which is 5 minutes per side with an additional 3 seconds for each move, becomes $5 \times 60 + 40 \times 3 = 420$ seconds. 420 seconds is less than 479 and is considered blitz. 40 comes from an approximation of the average number of moves in a game.

4.2 Metrics

The Mean Absolute Error (MAE) is used to provide an intuitive metric to understand the performance of models. The MAE is given for the pre-standardized rating to allow for a more interpretable loss. The test data is split up by time control to see how model performance varies between time controls. The metric used for the Chess Puzzle Competition is the mean squared error (MSE). We also provide the MSE for the game ratings.

5 Method

5.1 Input Features

We use a similar input representation as AlphaZero, with one plane for each piece-color combination, totaling 12 planes [26]. Each square contains a 1 if the corresponding piece is present; otherwise, it includes a 0. For example, the plane for black pawns contains 1s in the seventh row and 0s everywhere else. Following [25], the remaining clock-time is included as a feature. That paper showed a strong correlation between winning likelihood, remaining time, and position

¹ Games are taken after March 2021 because of some incorrect results that month in the Lichess database due to a datacenter fire

evaluation in 3-minute chess games. The remaining clock-time feature is standardized by subtracting the mean and dividing by the standard deviation. The same standardization process is applied to the ratings for more stable learning. The mean rating is 1514, with a standard deviation of 366. The same input is used for the puzzle competition, except it does not have clock time features because they were unavailable. FEN string representations were converted to the array representation.

5.2 Model Architecture

The architecture, which we will refer to as RatingNet, is a CNN that takes in each board state and passes this into a Bidirectional Long Short Term Memory Neural Network (LSTM) to predict both player ratings for the game after each move. CNNs have demonstrated success in various computer vision tasks, having been initially applied to document recognition [13]. LSTMs have also been proven to be successful in modeling long-term dependencies [9]. A four-layer CNN is used with batch normalization and mean pooling after each layer. The CNN is kept shallow to mitigate overfitting without having to train on a lot of data. The output of the CNN is concatenated with the amount of clock time spent for that corresponding move. This is then fed through the LSTM, which has two fully connected layers on top to output the ratings for the players of that game. The LSTM takes in the hidden state representation at the previous move and clock usage as input. One dropout layer is used after the first fully connected linear layer. Leaky ReLU activations are employed [32]. A simple baseline for predicting the mean chess rating is also shown for comparison.

RatingNet does not use one player’s rating as input to predict the other player’s rating. The reason is to focus the network on understanding chess rather than just predicting a rating close to the other player’s rating. This is to mitigate potential problems when the rating difference is significant, one player’s rating is provisional, or when a player is playing on an alternate account.

6 Results

Table 1. MAE of Different Methods Across Various Chess Time Controls

Time Control	RatingNet	RatingNetNoClock	Mean
Average Test Loss	182	239	346
Ultrabullet Loss	186	269	250
Bullet Loss	182	274	340
Blitz Loss	183	244	378
Rapid Loss	182	207	305
Classical Loss	151	185	246